



SK shieldus

Presentation by SK shieldus at CSK 2026

17 September 2026

Background

The background features a dark, deep blue gradient. In the lower half, there are several flowing, wavy lines in shades of orange and red, creating a sense of motion and depth. The lines are smooth and have a slight glow, contrasting with the dark background.

AI offensive capabilities have crossed a critical threshold

AI offensive capabilities, including autonomous zero-day discovery, have moved beyond theoretical limits and are now validated by real-world evidence

73%

Expert-level CTF success rate

- Claude Mythos Preview | UK AISI Evaluation (Apr 2026)¹⁾
- A level of complexity that remained effectively unsolved by any model prior to April 2025

2027

Autonomous discovery of a zero-day vulnerability in OpenBSD²⁾

- A 27-year-old denial-of-service vulnerability in the TCP SACK implementation
- Achieved with approximately 1,000 executions at a total cost of less than \$20,000

Assessment assumptions and considerations

Although the AISI cyber range did not simulate active defenders, the rise of autonomous AI-driven cyber threats is no longer in question

The collapse of the defense cycle: machine-speed attacks vs. human-speed defense

The fundamental breakdown of human-centric, manual patch management

Human-speed
(Defense)

Time to remediate high-risk vulnerabilities in enterprise applications and APIs

Average: 55 Days

Reversal of Time-to-Exploit (TTE): 63 Days → -7 Days (Compromised 7 Days Before Patch Release)

Machine-speed
(Attack)

Based on the Top 25%
of Threat Actors

[T+22 Seconds]

Initial access established and account takeover
completed using compromised credentials

[T+15 Minutes]

Internal network scanning and
lateral movement initiated

[T+ 72 Minutes]

Exfiltration of sensitive
data completed

Attacks are completed in little more than an hour, while defense (patching) still takes two months

The end of perimeter defense: the weaponization of identity

Attackers no longer hack in. They log in using stolen credentials

The transformation of the attack surface

65%

Initial access through compromised identities and credentials

90%

Identity- and credential-related security breaches

48%

Web browsers exploited as an attack vector

99%

Overprivileged cloud accounts

The evolution of state-sponsored cyber threats

FAMOUS CHOLLIMA

CrowdStrike designation · North Korea-linked

- A 130% year-over-year increase in activity
- Infiltration via employment fraud leveraging AI-generated deepfake identities
- Average dwell time of 122 days, enabling sustained access and sensitive data exfiltration

Identity, not the perimeter, has emerged as the new line of defense

Evolution of SOC and AI Adoption

SOC evolution timeline



AI adoption across SOC evolution stages

AI type	Core technologies	Key advantages	Key limitations
Rule-based Analytics	Signature- and threshold-based pattern matching	Simple to implement and easy to validate false positives	Limited effectiveness against emerging threats
Machine Learning	Anomaly detection and UEBA	Capable of detecting previously unknown threats through probabilistic analysis	Limited explainability and transparency
Generative AI Assistance	Natural language querying and automated reporting	Enhances junior analyst productivity and decision-making	Susceptible to hallucinations and factual inaccuracies
Agentic AI	Autonomous alert triage, multi-step reasoning, and autonomous response	Enables 24/7 autonomous security operations	Unclear governance and accountability frameworks

※ Source : Gartner Hype Cycle for Security Operations 2025; SANS SOC Survey 2024; ReliaQuest 2025

The Growing Alert Fatigue Crisis

3,000 Alerts a Day, 63% Left Unaddressed. The SOC Is at a Breaking Point

Average Daily Alert Volume

2,992

[Tool Sprawl]

Organizations manage an average of 70 security tools from 35 vendors, while fragmented alerts increase analytical complexity.

Percentage of Alerts Left Unaddressed

63%

[Configuration Gaps]

10% of SIEM rules remain non-functional due to parsing errors or missing log sources, leaving critical detection capabilities unavailable.

False Positive

83%

[Lack of Actionable Context]

Alerts alone are not enough. Without actionable context, analysts struggle to make accurate and timely decisions.

Mean Time to Acknowledge (MTTA)

56 min

[Analyst Burnout]

Nearly two-thirds of cybersecurity professionals report severe stress, while more than 83% of SOCs experience annual staff turnover.

Why Traditional SOC Operations Are Being Challenged: The Reality of Asymmetric Cyber Warfare

While attackers operate at machine speed through 24/7 autonomous AI agents, defenders continue to rely on human-driven security operations. Autonomous defense is the only sustainable way to eliminate this asymmetry

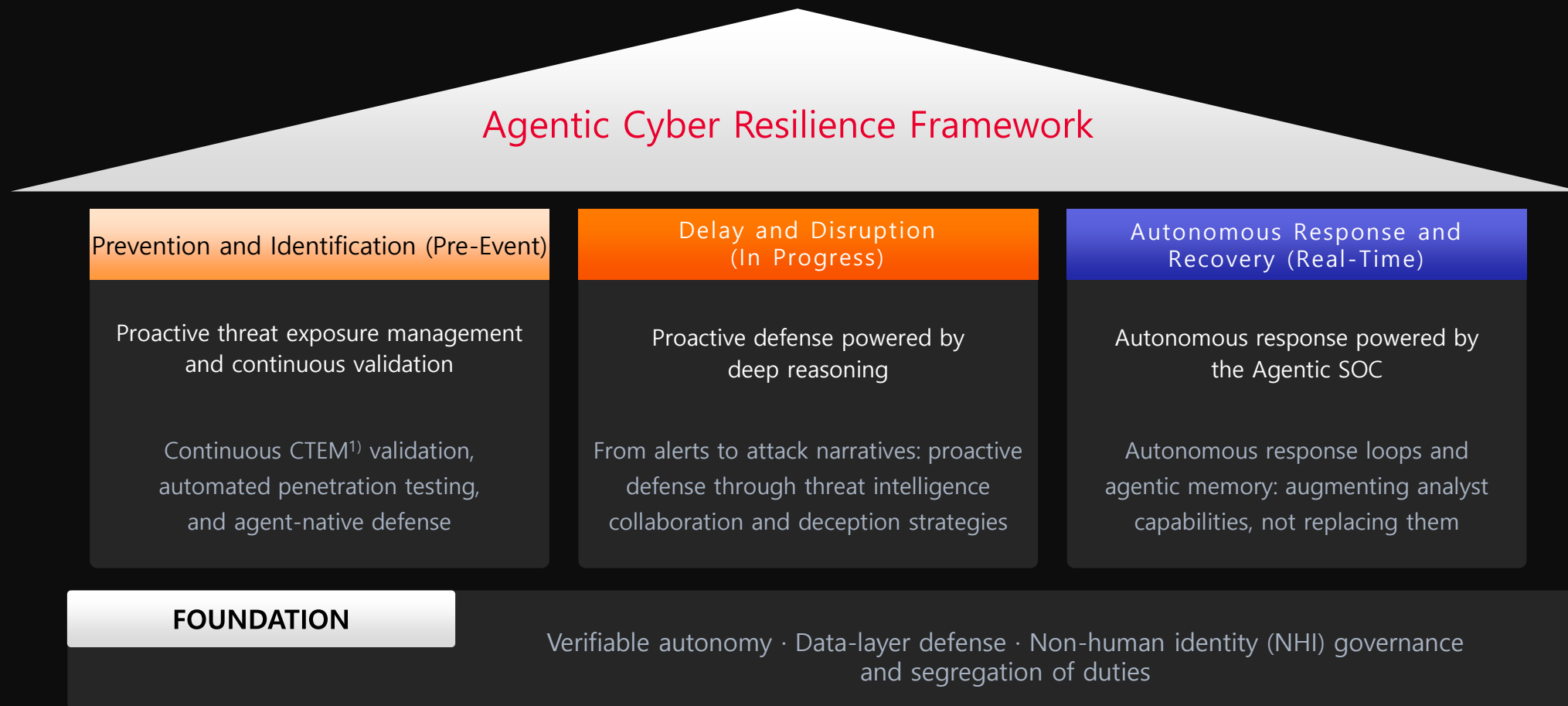
Threat velocity vs. response velocity	2m 7s from initial compromise to lateral movement	→	Deploying agentic MDR to enable autonomous detection and containment
Escalating alert volumes	3,000 daily alerts, 63% unaddressed	→	AI-driven alert triage, reducing analyst workload by up to 90%
Talent shortages and analyst burnout	4M talent shortage, 70% attrition risk	→	AI performs first-level analysis, while humans focus on decision-making and oversight
Growing operational complexity	70 security tools, 35 vendors, surging NHIs	→	Unified platforms powered by AI orchestration
Increasing compliance pressures	Growing regulatory pressure: EU AI Act and NIS2	→	Auditability and explainability for AI-driven decisions

Approach



A new defense paradigm: autonomous cyber resilience

The goal is no longer perfect prevention, but autonomous recovery without business disruption



1) CTEM(Continuous Threat Exposure Management) : A proactive security framework built on continuous threat exposure management

Proactive threat exposure management and dynamic validation

Continuously emulate adversary tactics through AI-driven penetration testing to identify Shadow AI and exploitable vulnerabilities before attackers do

Automated penetration testing

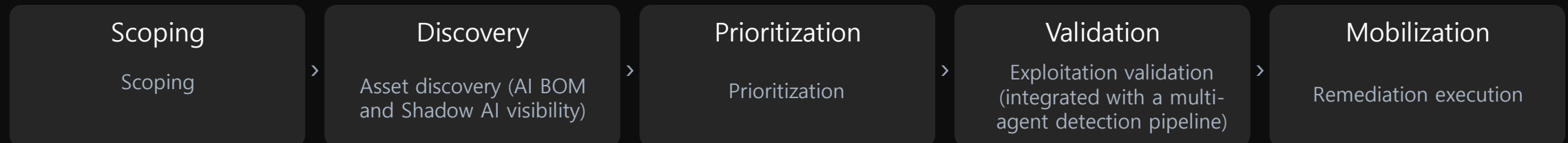
AS-IS One to two manual penetration tests per year, providing only a point-in-time snapshot.

TO-BE Continuously emulates real-world adversary tactics through AI to stress-test internal environments.

Continuous Threat Exposure Management (CTEM)

- MITRE ATT&CK¹⁾-based continuous validation
- Proactive identification of Shadow AI risks and misconfigurations
- AI BOM²⁾ -driven supply chain validation
- Prioritized remediation of actively exploited vulnerabilities

The 5 Stages³⁾ of Continuous Threat Exposure Management (CTEM)



1) MITRE ATT&CK: Adversary tactics and techniques based on real-world attacks 2) AI BOM: A complete inventory of AI system components
 3) Implement a Continuous Threat Exposure Management (CTEM) Program, Gartner(2022)

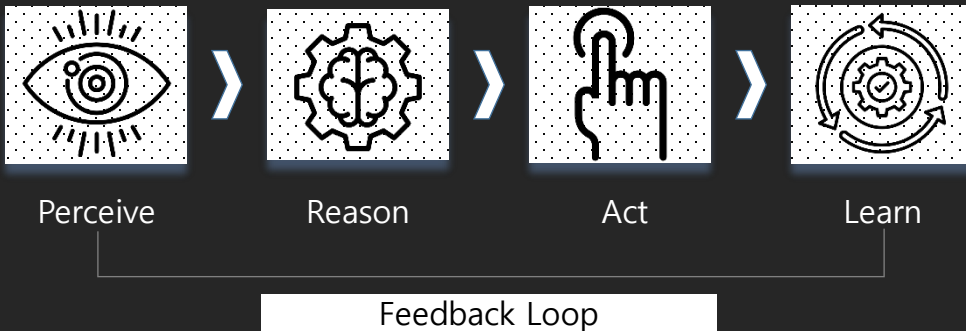
Agentic SOC: Definition and Architecture

An autonomous security operations architecture that perceives, reasons, acts, and learns

What Is an Agentic SOC?

Beyond traditional automation, an autonomous security operations architecture in which AI agents perceive, reason, act, and continuously learn from outcomes.

4 core characteristics of Agentic AI



Human-in-the-Loop

- Analyst dashboard with approval, rejection, and review workflows.
- Human approval for high-risk actions, with full auditing and feedback on agent decisions.

Orchestration Layer

- LLM-powered orchestration layer integrating playbook execution and SOAR capabilities.
- Dynamically coordinates agent activities and executes goal-driven, multi-step workflows.

Agent Layer

- Detection, investigation, response, and threat-hunting agents
- Each agent performs a specialized function, while multi-agent collaboration enables analysis of complex threats.

Data Layer

- SIEM, XDR, NDR, cloud telemetry, and threat intelligence
- Real-time collection and normalization of telemetry across endpoints, networks, cloud environments, and identities.

Response Layer

- Integrated with firewalls, EDR, and NAC to automate isolation, blocking, and patch deployment.
- Built-in capabilities for threat containment, account suspension, vulnerability remediation, and rollback.

Deep reasoning-driven proactive defense

Disrupting adversary toolchains to reduce the recurring cost of defending against the same threats

① From noise to narrative

AS-IS Event-level detection — Low-and-slow attacks misclassified as noise

TO-BE Correlating isolated events into a continuous attack narrative

Consolidates enterprise-wide workloads into a unified data lake, enabling reasoning agents to answer a critical question: "What is happening right now?"

Authorization does not imply legitimacy of behavior

② Proactive Defense: Four Strategic Pillars for the Korean Enterprise Environment

1

Disruption of adversary infrastructure (in coordination with law enforcement)
Partnering with KISA and law enforcement to disrupt malicious C2 infrastructure and stolen data channels

2

Threat Intelligence (TI)-Driven Exposure
Proactively analyze and share TTPs ¹⁾ and IoCs ²⁾ to disrupt the reuse of adversary tradecraft

3

Cyber Deception
Leverages decoy infrastructure and honeypots to consume adversaries' time and computational resources

4

Attack surface reduction and re-entry prevention
Automated remediation of attack paths and identity privileges

Proactive defense is not unlawful hack-back. It is the strategic disruption of adversary ecosystems through law enforcement collaboration and threat intelligence integration

FOUNDATION : Verifiable Agency

Dynamically governs every AI-driven decision, preventing autonomy from escalating into uncontrolled outcomes

Principle · What It Ensures

● Evidence

Automation without decision traceability cannot be audited

● Boundaries

Policy-defined boundaries between autonomous actions and human approval

● Accountability

Clearly defines the responsibilities and authority of every agent

Control Mechanisms

Sensitivity-driven dynamic access control

Dynamically adjusts runtime privileges based on data sensitivity

Segregation of Duties (SoD)

Prohibits end-to-end execution of critical actions by a single agent

Non-Human Identity (NHI)¹⁾ Governance and ITDR²⁾ Enforcement
Elimination of non-expiring API keys and behavioral baseline modeling

End-to-End Controls Input Injection Detection → Data Provenance Verification
→ API Tool Whitelisting → Output Data Loss Prevention & Audit Logging

1) NHI(Non-Human Identity) : Non-human identities, including API keys, service accounts, and AI agents

2) ITDR(Identity Threat Detection and Response) : A security framework that detects and responds to identity-related threats in real time.

AI-powered threat detection: QuITE technology for uncovering hidden attack patterns

Transforms irregular, low-signal alerts into a coherent attack narrative, fundamentally overcoming the limitations of conventional detection methods

AS-IS

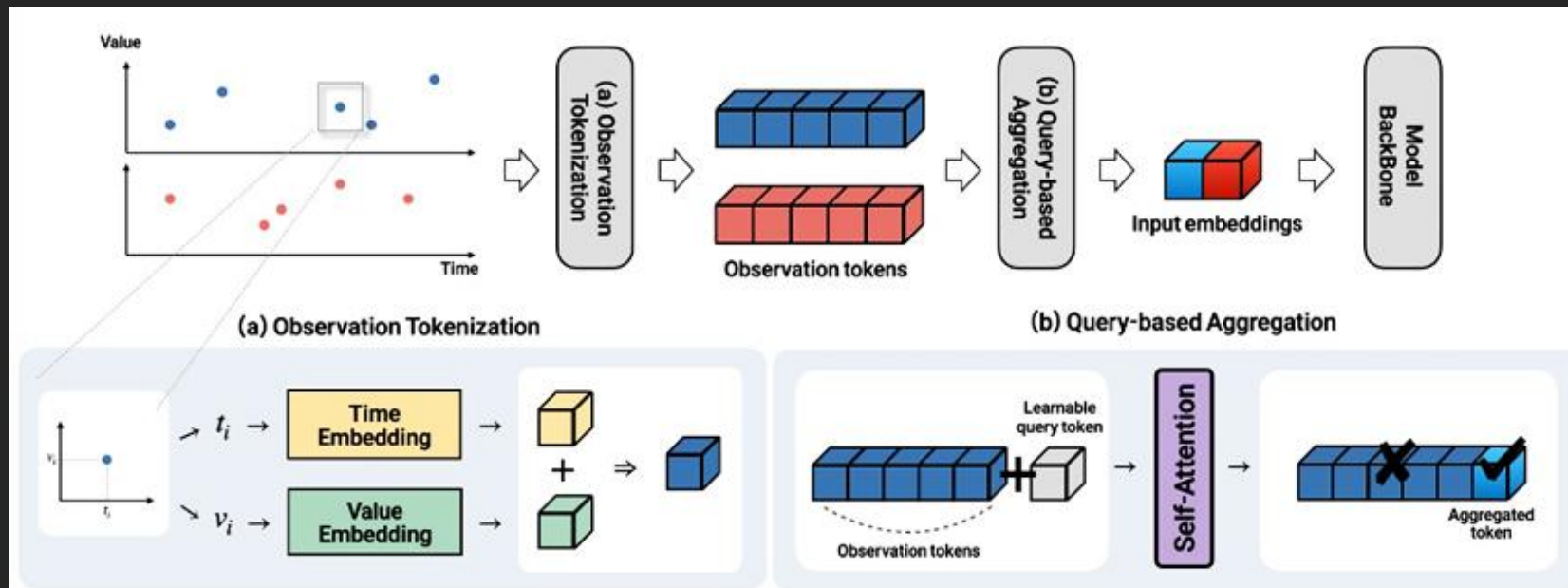
The trap of noise Stealthy low-and-slow attacks mistaken for noise

TO-BE

The evolution toward contextual reasoning

Reconstructing fragmented alerts into a continuous attack narrative

QuITE: Query-Based Irregular Time Series Embedding



Proprietary algorithm validated at ICML 2026, delivering a 54.7% improvement in predictive performance.

Outperforming conventional time-series analysis on global benchmark datasets

Multi-layered threat adjudication powered by a cybersecurity-specialized sLLM

Transforms AI reasoning and analyst feedback into institutionalized agentic memory, continuously enhancing detection and response quality

REASONING

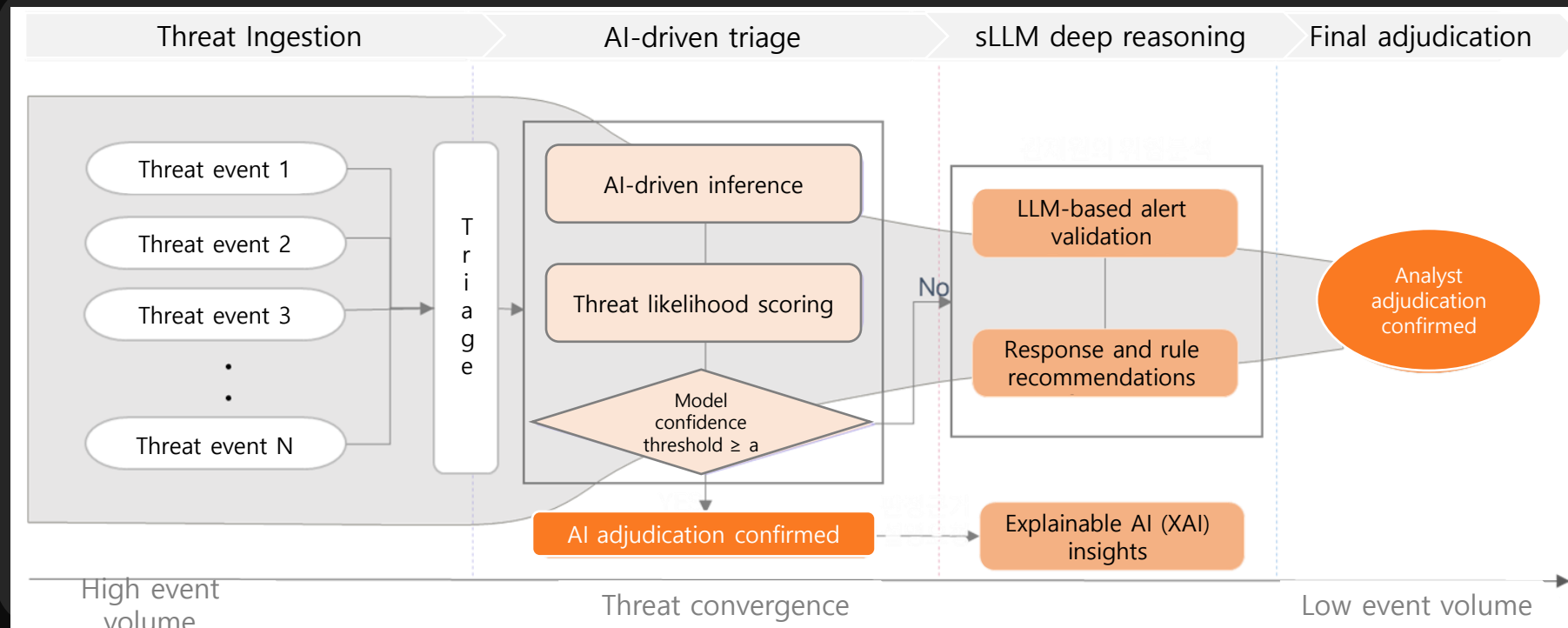
SK shieldus cybersecurity-specialized sLLM 1.0

HUMAN-AI TEAMING

Transparent threat rationale · Explainable AI (XAI)

AGENTIC MEMORY

Capturing human-AI collaboration outcomes as organizational knowledge assets



AI-powered threat adjudication and explainability

- Threat adjudication accuracy (deep learning): 95.0%
- Threat explainability (sLLM): 94.5%

Gartner Hype Cycle : AI SOC

Gartner has positioned AI SOC agents at the Peak of Inflated Expectations

Cybersecurity AI Assistant

Position: Peak of Inflated Expectations

42% of organizations have already deployed or piloted the technology, while an additional 46% plan to adopt it within the next year.

Agentic AI SOC

Position: Approaching the Peak of Inflated Expectations

Autonomously triaging alerts, investigating threats, and orchestrating responses, driving alert handling times down to 3-10 minutes.

CrowdStrike

Charlotte AI Agent (Agentic MDR)

- Delivers end-to-end autonomous security operations, spanning detection, investigation, and response.
- Leverages the Falcon Threat Graph to deliver autonomous, context-driven threat analysis.

Released in March 2026

Palo Alto Networks

XSIAM Autonomous Response

- Autonomous threat response workflows
- Provides AgentiX, an agentic platform for SOC automation.

Announced at RSAC 2026

Google Cloud

Convergence of SecOps and Wiz

- Security AI agents powered by Gemini
- A unified platform integrating cloud security with SOC operations

Planned for Release by the End of 2026

Microsoft

Copilot for Security (with autonomous response)

- Transitioning from security assistants to autonomous agents
- Enables automated security operations through Microsoft Sentinel integration.

In Development (2026)

Agentic SOC evaluation metrics

The success of an Agentic SOC must be measured through clearly defined performance metrics



Gartner vendor assessment

- Ability to Execute
- Completeness of Vision
- Breadth of AI SOC agent coverage

Alignment with Gartner Magic Quadrant and Hype Cycle



Technical performance metrics

- MTTD (Mean Time to Detect)
- MTTR (Mean Time to Respond)
- False Positive Rate (FPR) and False Negative Rate (FNR)

Measured against the internal baseline throughout the pilot period



Compliance

- Compliance with Korean cybersecurity and data protection regulations
- Alignment with PIPA and ISMS-P requirements
- Support for AI decision auditability and traceability

Key requirements for public sector customers



Total cost of ownership (TCO)

- Licensing, implementation, and operational costs
- Analyst productivity and workforce efficiency gains
- Cost avoidance through breach prevention and risk reduction

Three-year ROI analysis



Operational resilience

- Rollback capabilities for autonomous response failures
- Human-in-the-loop support
- Multi-tenancy support

Criteria for minimizing impact on customer environments

Key takeaways

As autonomous cyber defense advances, human operators evolve from responders to cybersecurity orchestrators



Machine-speed defense for machine-speed threats

As cyber attacks operate at machine speed, autonomous defense becomes not a choice, but a condition for survival.



From operator to orchestrator

The role of the security analyst is shifting from alert processing to orchestrating cyber defense through high-stakes decision-making and AI agent governance.



Trust enables autonomy

The Agentic SOC becomes viable only through transparent AI decisions, assured rollback capabilities, and human-in-the-loop control.

Pioneering the most intelligent and trusted cybersecurity paradigm through an Agentic AI SOC

Partnering with SK shieldus to build the future of autonomous cyber resilience

Appendix



From reasoning AI to acting AI: the collapse of trust boundaries

As agents actively interact with external tools, the attack surface expands exponentially, significantly increasing the scope of what must be defended

Past · Static AI

Inference Tool

Limited to cognitive tasks such as answering questions and classifying data.

Defense focus

Focused on model integrity and bias mitigation



Present · Agentic AI

Autonomous Action

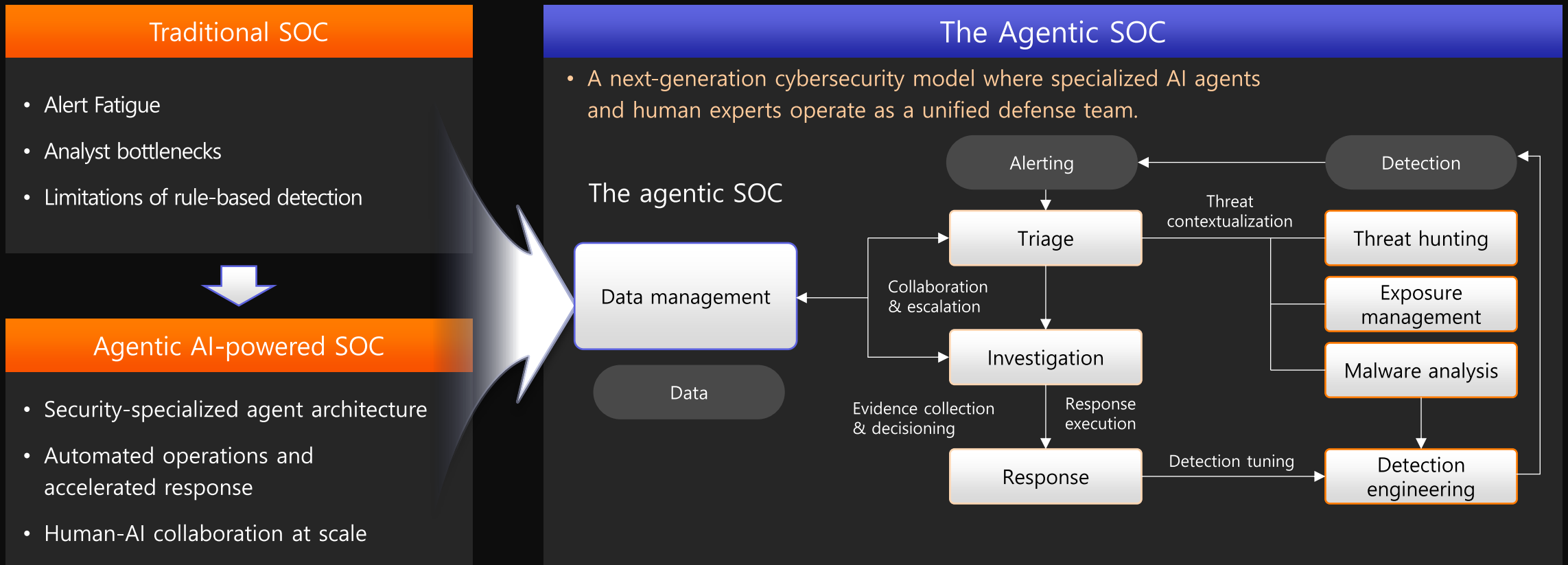
Autonomous agents that interact with their environment, invoke external tools, and take action independently.

Key risks

The collapse of trust boundaries and the exponential expansion of the attack surface

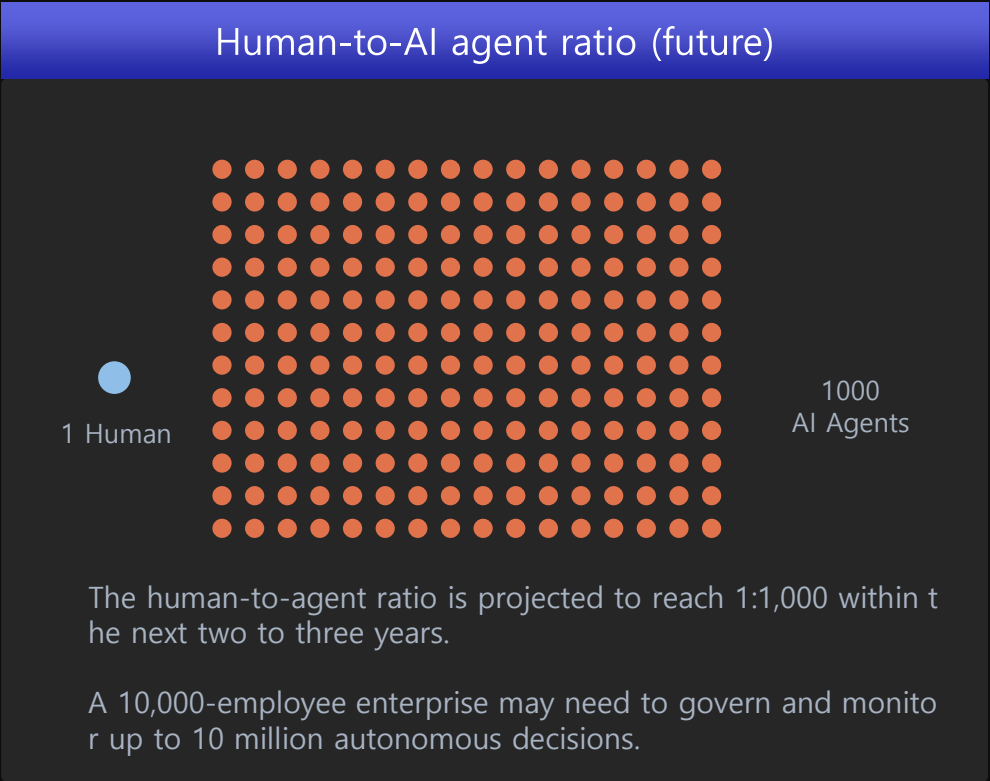
Agentic SOC-driven autonomous response

As threats are contained autonomously at machine speed, security analysts evolve into commanders who govern, validate, and direct cyber defense operations



The Agentic AI paradox: one defender versus a thousand autonomous attackers

The emergence of millions of autonomous agents will soon push traditional, human-centric security controls beyond their limits



Asymmetry matrix		
	Adversary (AI)	Defender (Human)
Speed	Machine-speed automation Attacks executed within minutes	Human intervention and manual analysis Ticket processing delays
Scale	Enterprise-scale network reach Multi-target simultaneous attacks	Fragmented and disconnected visibility
Operating model	Continuous learning and adaptation	Reactive, rule-based detection

This is not a gap that can be closed by adding more personnel — it requires a fundamental shift in the defense model.

* Source : How to Reshape Defense Faster than Attackers Reshape Offense, RSAC2026

The Journey Toward Autonomous Cyber Resilience

Strategy, technology, and execution partner



SECUDIUM: a next-generation platform for integrated security operations

Transitioning to an autonomous defense model that combines machine-speed operations, explainable AI, and unparalleled security operations data

SECUDIUM MXDR

17 billion

Events ingested per day

75%

Reduction in AI-powered threat adjudication time

110,000

Average daily AI-powered threat detection and response

Roadmap to the Agentic SOC

[Current]

AI-powered threat adjudication

- AI-assisted adjudication with analyst approval
- SOAR playbook orchestration
- Explainable threat adjudication via sLLM



[Next 1-2 years]

Agentic SOC transformation phase

- Advanced AI/ML and sLLM-powered threat adjudication
- Core Agentic AI adoption
- Human-in-the-loop governance



[Next 2-3 years]

Agentic SOC maturity phase

- Autonomous multi-agent operations
- 24/7 autonomous threat detection and response
- Continuous learning and feedback loops

Evolving toward an autonomous defense architecture powered by 24/7 multi-agent detection, analysis, and response

Why Agentic SOC?

Cyber threats now operate at a velocity beyond human perception and response
 "Autonomous agents are now a cybersecurity imperative" – RSAC 2026

Accelerating Attack Velocity

2m 7s

Initial Compromise → Lateral Movement

The zero-day weaponization window has shrunk from 771 days in 2018 to less than one hour in 2026.

Inherent Limitations

63%

2,992 Alerts Left Unaddressed Daily

The surge in security alerts has outpaced human capacity, with 66% of SOC analysts unable to keep up.

The Rise of Weaponized AI Agents

24/7

Autonomous Offensive Agents

Attackers already operate with autonomous AI agents. Defenders can no longer compete without them.

The Weaponization of AI Agents

Claude Mythos: AI as a Cyber Weapon

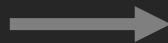
Key takeaways from the Claude Mythos preview

- Announcement Date: April 7, 2026 (Anthropic)
- Core Capability: Autonomous zero-day discovery and exploit generation (83%)
- Restricted Disclosure: Dramatic Increase in the Potential for Cyber Attack Automation
- Project Glasswing: Limited early access for trusted institutions
- 2025 Precedent: AI-orchestrated cyber espionage campaign by a state-sponsored Chinese threat group

The collapse of traditional zero-day defenses

[2018 Average]

771 Days



[2026 Average]

Within 1 hour



Impact on SOC Operations

- 1** Dramatic compression of attack timelines
The zero-day weaponization window has collapsed from days to less than one hour.
- 2** The weaponization of open-source LLMs
Comparable open-source sLLMs are already being fine-tuned and weaponized for cyber operations.
- 3** 24/7 autonomous offensive agents
Continuously identifying vulnerabilities and refining attack chains without interruption.
- 4** Cyber losses have reached record levels
Average breach cost: \$4.44M globally (\$10.22M in the U.S.)
- 5** Surging non-human identity (NHI) threats
AI agents and API keys now outnumber human identities by 80 to 1.

How AI has reshaped the threat landscape

Attack capabilities have crossed a critical threshold,
as demonstrated by real-world evidence



Agentic SOC evaluation metrics

The Agentic SOC fundamentally overcomes the limitations of third-generation SOC

	Third-generation SOC (current)	Fourth-generation SOC (Agentic SOC)
Alert handling	Analyst-driven manual triage, with an average wait time of 56 minutes	Autonomous AI agent triage (resolved within 3-10 minutes)
Investigation approach	Analysts manually navigate multiple security tools for investigation	Multi-agent autonomous investigation with automated collection of relevant artifacts
Response speed	Decision → Approval → Execution (tens of minutes to hours)	Autonomous containment and blocking within seconds to minutes after threat identification
Operational coverage	Shift-based operations (limited resilience during nights and holidays)	24/7/365 uninterrupted autonomous operations
Scalability	Dependent on workforce expansion, with clear cost constraints	Elastic scalability through the deployment of additional agents
Human role	Analysts perform first-level review of every alert (severe burnout and attrition risk)	High-impact decision-making, oversight, and guardrail governance (cybersecurity orchestrator)
Learning and optimization	Quarterly rule optimization with slow improvement cycles	Continuous learning through feedback loops with autonomous adaptation to emerging threat patterns

Agentic SOC adoption strategy

The adoption of an Agentic SOC is not a technology initiative, but a fundamental shift in operating philosophy

Successful adoption of an Agentic SOC requires more than deploying technology. It demands a strategic five-step approach spanning data quality, governance, and continuous improvement

1

Data quality and readiness

Validate log collection, normalization, and labeling quality, while remediating SIEM normalization rule errors, which average 18%

2

Pilot implementation

Begin with high-volume, repetitive use cases such as WAF false-positive tuning and Tier-1 EDR triage, while establishing clear success metrics including MTDD and MTTR

3

Human-in-the-loop design

- Clearly define the boundaries of autonomous response, distinguishing actions that can be automated from those requiring human approval
- Implement rollback mechanisms to ensure rapid recovery from unintended actions

4

Establishing governance

- Mandate comprehensive audit trails for all AI-driven decisions
- Enforce least-privilege access controls for every agent

5

Continuous Improvement

Drive continuous optimization through model retraining based on operational feedback, ongoing alignment with MITRE ATT&CK, and regular Red Team assessments